

# longops

Long operations and detailed calculations.  
Opérations posées et calculs détaillés.

v0.1.0

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## About

As a mathematics teacher, I sometimes need to show how to carry out arithmetic “by hand”. I am now making available the functions that I originally created for my own use.

I previously used the LaTeX package **xlop**, and since Typst is much more convenient to use, I decided to create my own equivalent. This package is **not** a direct translation of **xlop**: it includes some features that **xlop** does not, while omitting others, according to my own needs.

I occasionally used AI tools (mainly Gemini for the **Order of Operations** module, and also to add features to the other functions that I had initially implemented for just one of them).

→ All the examples are either in the README or the french manual

## Functions for handwritten arithmetic layouts ...

- `add-en-ligne()`
- `addition()`
- `division()`
- `egalite-decimale()`
- `egalite-euclidienne()`
- `multiplication()`
- `racine()`
- `sous-en-ligne()`
- `soustraction()`

### add-en-ligne

Compute the sum of all the supplied numbers inline.

#### Parameters

`add-en-ligne(..ops)` -> `content`

### addition

Lay out the column addition of 2 or more numbers with the possibility of hiding the result with `hide-result`, or a liste of some of the digits and then to show the solution, a few colors can be set

#### Examples :

## Parameters

```
addition(  
  show-carry: boolean,  
  couleur: color,  
  size: length,  
  fr: boolean,  
  hide-result: boolean,  
  couleur-cadre: color,  
  couleur-solution: color,  
  type-mask: str,  
  liste: array,  
  solution: boolean,  
  signe: boolean content,  
  space: length,  
  texte: length,  
  ..args  
) -> content
```

**show-carry** boolean

show or hide the carries

Default: **true**

**couleur** color

color of the carries

Default: red

**size** length

Size of the columns

Default: **1em**

**fr** boolean

french decimal separaro “,” or not

Default: **true**

**hide-result** boolean

Hide the carry row and the result

Default: **false**

**couleur-cadre** `color`

Color of the mask of hidden digits

Default: `blue.mix(gray)`

**couleur-solution** `color`

Color of the hidden digits if solution = true

Default: `red`

**type-mask** `str`

How the digits are hidden : "rect" for a round rectangle or "." for dots

Default: `"rect"`

**liste** `array`

list of the digits to hide

Default: `()`

**solution** `boolean`

show the carries and the hidden digits

Default: `false`

**signe** `boolean` or `content`

Sign of equality, false or a symbol \$ = \$

Default: `false`

**space** `length`

Space if more than one calculation on a line, default : 1fr

Default: `1fr`

**texte** `length`

size of the text : 1em

Default: `1em`

## division

Divide decimal number a by b.

With extradigits, computes additional digits after the decimal point beyond those already present in a (when cycle:false). These extra digits are displayed in gray (mode:"g" by default), white (mode:"0"), as dots (mode:"p" or "gp"), or omitted (mode:""). Use s:true/false to show or hide the subtraction steps. You can highlight the repeating cycle in the quotient using cycle:true (underlined with cycle-color), display it in parentheses with cycle:"", or display it in color with cycle:red. Fill-in-the-blank divisions are also supported with hide-result, liste, and solution (using the type-mask, couleur-cadre, and couleur-solution parameters).

## Examples:

### Parameters

```
division(  
  a: int float ,  
  b: int float ,  
  mode: str ,  
  extradigits: int ,  
  s: boolean ,  
  cycle: boolean str color ,  
  cycle-color: color ,  
  fr: boolean ,  
  size: length ,  
  dx,  
  dy,  
  hide-result: boolean ,  
  liste: array ,  
  solution: boolean ,  
  type-mask: str ,  
  couleur-cadre: color ,  
  couleur-solution: color ,  
  space: length  
) -> content
```

**mode**    str

Display mode for zeros added to the dividend: "g" (gray), "0" (black), "p" (small dot), "gp" (large dot), "" (empty)

Default: "g"

**extradigits**    int

Number of additional quotient digits to compute

Default: 0

**s**    boolean

Show or hide the subtraction steps

Default: true

**cycle**    `boolean` or `str` or `color`

Detect and display the repeating cycle automatically

Default: `false`

**cycle-color**    `color`

Color of the cycle or underline

Default: `luma(50%)`

**fr**    `boolean`

If `fr:true`, the decimal separator is “,” otherwise “.”

Default: `true`

**size**    `length`

Column width

Default: `.7em`

**dx**

dx for the `)` in mode `fr:false`

Default: `.6em`

**dy**

dy for the `)` in mode `fr:false`

Default: `-.82em`

**hide-result**    `boolean`

Hide the working steps and quotient (exercise mode)

Default: `false`

**liste**    `array`

List of cell indices to hide (dividend, divisor, steps, quotient)

Default: `()`

**solution**    `boolean`

Show or hide the solution inside the masks

Default: `false`

**type-mask**    `str`

Mask type ("rect" or "line")

Default: `"rect"`

**couleur-cadre**    `color`

Color of the mask border

Default: `blue.mix(gray)`

**couleur-solution**    `color`

Color of the solution text

Default: `red`

**space**    `length`

Spacing when several calculations are on the same line

Default: `1fr`

## egalite-decimale

Write the equality corresponding to a decimal division.

### Parameters

```
egalite-decimale(  
  a: int float ,  
  b: int float ,  
  s: str  
) -> content
```

## egalite-euclidienne

Write the equality corresponding to a Euclidean division.

## Parameters

```
egalite-euclidienne(  
  a: int float ,  
  b: int float  
) -> content
```

## multiplication

Lay out the multiplication of 2 integer or decimal numbers with the possibility of hiding intermediate rows using hide-result, hiding a liste of selected digits, then displaying the solution, with various configurable colors.

## Examples:

### Parameters

```
multiplication(  
  a: int float ,  
  b: int float ,  
  mode: str ,  
  sym: boolean str ,  
  fr: boolean str ,  
  size: length ,  
  hide-result: boolean ,  
  liste: array ,  
  solution: boolean ,  
  type-mask: str ,  
  couleur-cadre: color ,  
  couleur-solution: color ,  
  space: length  
) -> content
```

**mode** str

Display mode for shifted rows: “g” (default) for gray zeros, “0” for normal zeros, “p” for small dots, “gp” for larger dots, “” or anything else for nothing

Default: "g"

**sym** boolean or str

Show the addition “+” signs and the final “=” if true; only “+” if “+”, only “=” if “=”, otherwise nothing

Default: true

**fr** boolean or str

If fr:true, the decimal separator is “,” otherwise “.”

Default: true

**size**    length

Column width

Default: 1em

**hide-result**    boolean

Hide intermediate rows and the final result for exercises

Default: false

**liste**    array

List of cell indices to hide (fill-in-the-blank multiplication)

Default: ()

**solution**    boolean

Show or hide the solution inside the masks

Default: false

**type-mask**    str

Mask type ("rect" or "line")

Default: "rect"

**couleur-cadre**    color

Color of the mask border

Default: blue.mix(gray)

**couleur-solution**    color

Color of the solution text

Default: red

**space**    length

Spacing when several calculations are on the same line

Default: 1fr

## racine

Compute the square root of a, with extradigits additional decimal places, optionally showing digit pairs using groupes in couleur-groupes, displaying the construction step by step with step, hiding subtraction steps with s, and hiding the result or a liste of digits.

### Examples:

#### Parameters

```
racine(  
  a: int float ,  
  extradigits,  
  groupes,  
  couleur-groupes,  
  step,  
  couleur-step,  
  s,  
  fr: boolean ,  
  size: length ,  
  hide-result: boolean ,  
  liste: array ,  
  solution: boolean ,  
  type: str ,  
  couleur-cadre: color ,  
  couleur-solution: color ,  
  space,  
  ..args  
) -> content
```

**fr** boolean

If fr:true, the decimal separator is “,” otherwise “.”

Default: **true**

**size** length

Column width

Default: **0.65em**

**hide-result** boolean

Hide the construction steps, operations, and result (exercise mode)

Default: **false**

**liste** array

List of cell indices to hide (fill-in-the-blank exercise)

Default: **()**

**solution**    `boolean`

Show or hide the solution inside the masks

Default: `false`

**type**    `str`

Mask type ("rect" or "line")

Default: `"rect"`

**couleur-cadre**    `color`

Color of the mask border

Default: `blue.mix(gray)`

**couleur-solution**    `color`

Color of the solution text

Default: `red`

### sous-en-ligne

Compute the difference of all the supplied numbers inline.

#### Parameters

```
sous-en-ligne(  
  ..ops,  
  digits: auto int  
) -> content
```

### soustraction

Lay out the subtraction of 2 or more numbers with the possibility of hiding the result row with `hide-result`, hiding a liste of selected digits, then displaying the solution, with various configurable colors.

#### Examples:

## Parameters

```
soustraction(  
  show-borrow: boolean ,  
  barre: boolean ,  
  fr: boolean ,  
  couleur: color ,  
  size: length ,  
  zeros: boolean ,  
  hide-result: boolean ,  
  couleur-cadre: color ,  
  couleur-solution: color ,  
  type-mask: str ,  
  liste: array ,  
  solution: boolean ,  
  signe: boolean content ,  
  texte: length ,  
  space: length ,  
  ..args: array  
) -> content
```

**show-borrow**    boolean

Show or hide the borrows

Default: **true**

**barre**    boolean

If fr:false, show or hide the crossed-out original digit

Default: **false**

**fr**    boolean

French notation or English borrowing method

Default: **true**

**couleur**    color

Color of the borrows

Default: red

**size**    length

Column width

Default: **1em**

**zeros**    `boolean`

Add trailing zeros when needed

Default: `true`

**hide-result**    `boolean`

Hide the borrows and the result

Default: `false`

**couleur-cadre**    `color`

Color of the border around hidden digits

Default: `blue.mix(gray)`

**couleur-solution**    `color`

Color of hidden digits when solution = true

Default: `red`

**type-mask**    `str`

How hidden digits are displayed: “rect” for a rounded rectangle or ...

Default: `"rect"`

**liste**    `array`

List of digits to hide in a fill-in-the-blank subtraction

Default: `()`

**solution**    `boolean`

Show the borrows and the result in hide-result mode (for answer keys)

Default: `false`

**signe**    `boolean` or `content`

Equality sign, false or a symbol \$ = \$

Default: `false`

**texte**    `length`

Text size: 1em

Default: `1em`

**space**    `length`

Spacing when several calculations are on the same line

Default: `1fr`

## Step-by-step calculations and highlighting

- `detail()`
- `etapes-calcul()`

### detail

Function for writing calculation steps manually.

#### Parameters

```
detail(  
  expr-str: str,  
  vertical: boolean,  
  arrondi: boolean,  
  digits: int,  
  appro: int,  
  todo: boolean,  
  space: length,  
  a: color,  
  b: color,  
  f: color,  
  g: color,  
  r: color,  
  l: color,  
  m: color,  
  n: color,  
  o: color,  
  t: color,  
  p: color,  
  y: color,  
  c: str  
) -> content
```

**expr-str**    `str`

The calculation expression as described above

**vertical**    `boolean`

Vertical layout

Default: `false`

**arrondi** `boolean`

Append a rounded value at the end

Default: `false`

**digits** `int`

Number of decimal places for the rounded value

Default: `2`

**appro** `int`

Step from which “=” is replaced by “≈” when appropriate

Default: `0`

**todo** `boolean`

If `todo: true`, only the initial expression is displayed

Default: `false`

**space** `length`

Spacing before the next calculation

Default: `1fr`

**a** `color`

a: aqua; can be changed from Typst’s default color if needed

Default: aqua

**b** `color`

b: blue; can be changed from Typst’s default color if needed

Default: blue

**f** `color`

f: fuchsia; can be changed from Typst’s default color if needed

Default: fuchsia

**g** 

g: green; can be changed from Typst's default color if needed

Default: green

**r** 

r: red; can be changed from Typst's default color if needed

Default: red

**l** 

l: gray; can be changed from Typst's default color if needed

Default: gray

**m** 

m: maroon; can be changed from Typst's default color if needed

Default: maroon

**n** 

n: rgb(8,111,189) (navy); can be changed if needed

Default: `rgb( "#086fbd" )`

**o** 

o: orange; can be changed from Typst's default color if needed

Default: orange

**t** 

t: rgb(0,128,128) (teal); can be changed if needed

Default: `rgb( 0, 128, 128 )`

**p** 

p: purple; can be changed from Typst's default color if needed

Default: purple

**y**   `color`

y: yellow; can be changed from Typst's default color if needed

Default: yellow

**c**   `str`

Display mode: "cancel", "circle", or "color"

Default: "color"

## etapes-calcul

Show the step-by-step evaluation of `expr-str` (unless `todo:true`), with the option to perform independent operations of the same precedence simultaneously using `concomitant`, display the work vertically or horizontally, highlight the current step (`highlight`), and use fraction mode to keep computations exact by forcing calculations to continue with fractions.

### Parameters

```
etapes-calcul(  
  expr-str: str,  
  mode: str,  
  concomitant: boolean,  
  vertical: boolean,  
  digits: int,  
  n: int,  
  name: str content,  
  todo: boolean,  
  space: length,  
  fraction: boolean,  
  highlight: color boolean none,  
  skip: array  
) -> content
```

**mode**   `str`

"fr" or "en" mode for the decimal separator

Default: "fr"

**concomitant**   `boolean`

Set to true to perform independent operations of the same precedence simultaneously

Default: false

**vertical**    `boolean`

Display the calculation vertically

Default: `false`

**digits**    `int`

Number of decimal places for calculations involving  $\pi$ , etc.

Default: `2`

**n**    `int`

Number of steps (counted from the end) at which “=” is replaced by “ $\approx$ ” when appropriate

Default: `0`

**name**    `str` or `content`

Label to prepend to the expression

Default: `""`

**todo**    `boolean`

Exercise mode: only the initial expression followed by “=” is displayed

Default: `false`

**space**    `length`

Horizontal spacing between two etapes-calcul outputs

Default: `1fr`

**fraction**    `boolean`

In fraction mode, computations remain exact, either as fractions or simplified forms such as  $5\pi$

Default: `false`

**highlight**    `color` or `boolean` or `none`

Either a color used to highlight the current operation, `false` for bold, or `none` for no highlighting

Default: `rgb("#1e88e5")`

**skip** array

List of steps to skip if needed

Default: ()